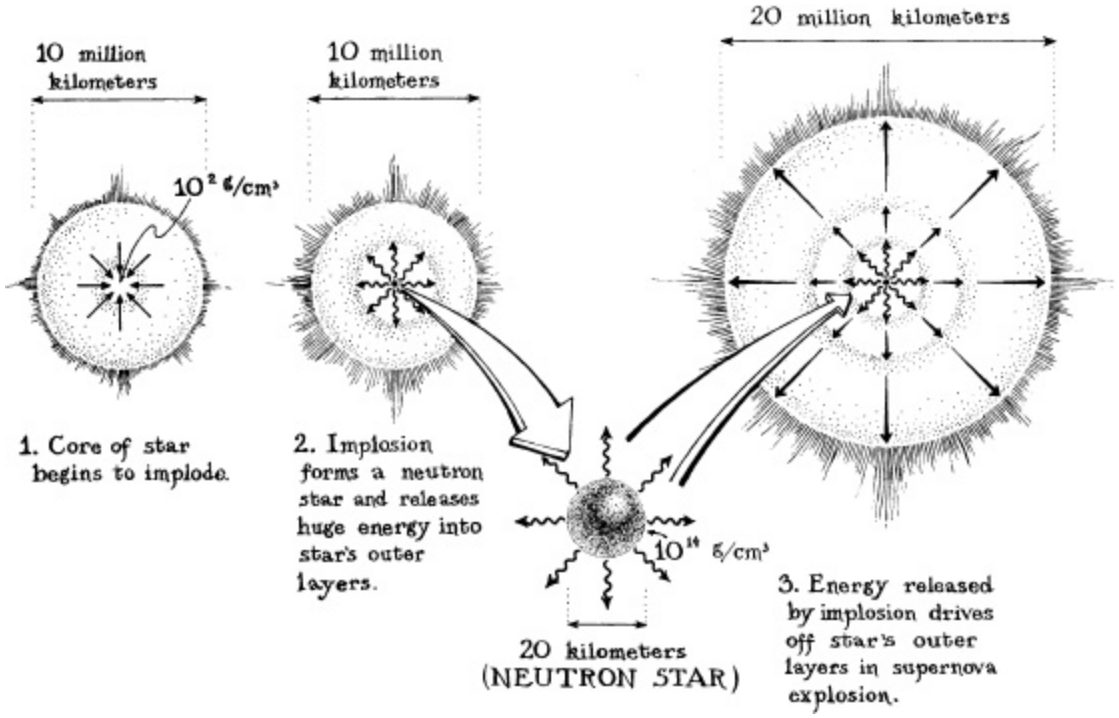
Zwicky did not know what might initiate the implosion of the star's core and convert it into a neutron star, nor did he know how the core might behave as it imploded, and therefore he could not estimate how long the implosion would take (is it a slow contraction or a high-speed implosion?). (When the full details were ultimately worked out in the 1960s and later, the core turned out to implode violently; its own intense gravity drives it to implode from about the size of the Earth to 100 kilometers circumference in less than 10 seconds.) Zwicky also did not understand in detail how the energy from the core's shrinkage might create a supernova explosion, or why the debris of the explosion would shine very brightly for a few days and remain quite bright for a few months, rather than a few seconds or hours or years. However, he knew-or he thought he knew-that the energy released by forming a neutron star was the right amount, and that was enough for him.



5.1 Fritz Zwicky's hypothesis for triggering supernova explosions: The supernova's explosive energy comes from the implosion of a star's normal-density core to form a neutron star.

Box 5.2

The Equivalence of Mass and Energy